

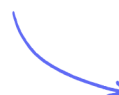
Calculator Questions

Algebraic Roots & Indices

Algebraic Roots & Indices

Easy (3 questions)	/6
Medium (5 questions)	/14
Hard (3 questions)	/7
Very Hard (5 questions)	/12
Total Marks	/39

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Easy Questions

1 Simplify $a^2 \div a^6$.

(1 mark)

2 (a) Simplify $p^2 \times p^4$.

(1 mark)

(b) Simplify $m^{15} \div m^5$.

(1 mark)

(c) Simplify $(k^3)^5$.

(1 mark)

3 Simplify.

$$5m^2 \times 2m^3$$

(2 marks)

Medium Questions

1 $2^{3x+1} = 8^{2-x}$

Find the value of x

$x = \dots\dots\dots$

(3 marks)

2 Simplify.

$$(x^8y^7) \div (x^{-1}y^3)$$

(2 marks)

3 Simplify.

$$(27x^9)^{\frac{2}{3}}$$

(2 marks)

4 Simplify $(81y^{16})^{\frac{3}{4}}$.

(2 marks)

5 (a) Simplify $(3p^2)^5$.

(2 marks)

(b) Simplify $18x^2y^6 \div 2xy^2$.

(2 marks)

(c) Simplify $\left(\frac{5}{m}\right)^{-2}$.

(1 mark)

Hard Questions

1 Simplify.

$$\left(\frac{x^3}{8}\right)^{-\frac{4}{3}}$$

(2 marks)

2 Simplify $\sqrt[3]{27t^{27}}$.

(2 marks)

3 Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3 marks)

Very Hard Questions

1 Simplify $(256x^{256})^{\frac{3}{8}}$.

(3 marks)

2 $\sqrt[3]{y^2} = \sqrt[6]{x}$ and $y = \sqrt[n]{x}$.

Find the value of n .

$n = \dots\dots\dots$

(2 marks)

3 $x^{-\frac{1}{3}} = 32x^{-2}$

Find the value of x .

$x = \dots\dots\dots$

(3 marks)

4 Simplify $\left(\frac{16a^8}{c^{12}}\right)^{\frac{3}{4}}$.

(2 marks)

5 Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

find the value of m

$m = \dots\dots\dots$

(2 marks)